



Estimate Stiffness

The four flex numbers of any EI curve -- yours, a target, a competitor's -- between two contact points.

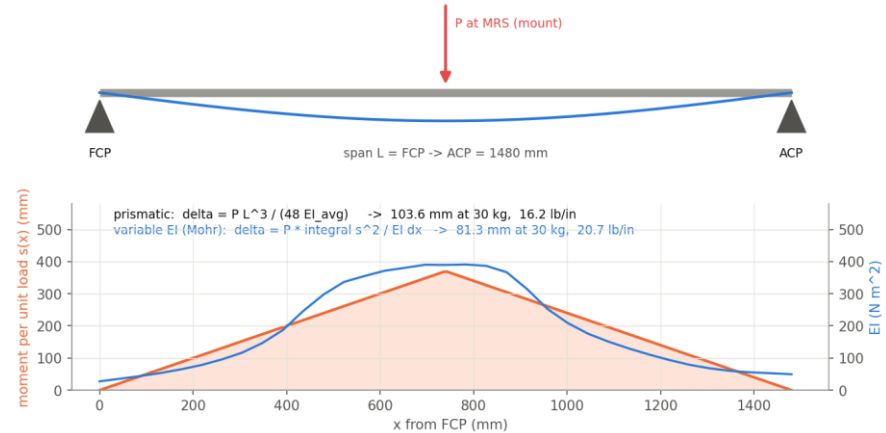


What it does

EI and Cross Section gives flex numbers for a modelled layup. Estimate Stiffness gives the same four numbers for any EI curve, so a measured or target curve can be compared on the same scale.

Use it when

- Rate a target or measured EI curve
- Compare two EI curves by the numbers
- Check a hand-edited EI profile



The test behind the numbers: FCP-ACP simply supported, load at the mount.



The dialog

Estimate Stiffness 1

✓

✗

1

El Edges
Edge of Measured_EI

(Q)

2

FCP

(Q)

3

ACP

(Q)

4

> Stiffness estimates

5

Recalculate

Final

#	Parameter	What it means
1	El Edges	The EI curve: an edge in the XZ plane, world Z in mm = EI in N m ² (EI and Cross Section's <name>_EI, a measured or a target curve).
2	FCP	Front contact point (X only).
3	ACP	Aft contact point (X only).
4	Stiffness estimates	Read-only: prismatic lb/in and mm/30 kg; estimated (variable EI) lb/in and mm/30 kg.
5	Recalculate	Button: compute the numbers.



Tips and limits



Dialog only

Numbers are computed when you press Recalculate and do not follow later geometry changes.



FCP before ACP

If FCP is not before ACP in X nothing is computed, silently.



Short curves

An EI curve shorter than FCP-ACP is extended to reach them.



Where it lives

Document	xSection (f8deedeb1fbd819a8fa20113), 'Stiffness Tools'
Tab	estimateStiffness (+ beam/xSectBeamAnalysis.fs)
Tests	none
Examples	Test studio: Estimate Stiffness 1
Explainer	docs/explainers/xsection/xsection_explained.md, section 1.6, 2.3
Related	EI and Cross Section, Estimate Stiffness, Estimate Deflection, Generate / Analyze baseline, Update profile